

MELANIE REBECCA

Senior Software Engineer | Backend & Full Stack | Node.js • TypeScript • React • AI-Assisted Development
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Based in Dubai, UAE | Open to Remote Opportunities Worldwide | Open to Relocation across Europe (Visa Sponsorship Required)

PROFESSIONAL SUMMARY

Senior Full Stack Engineer with 8+ years of experience designing scalable backend systems for e-commerce, payments, logistics, and insurance platforms. Specialized in Node.js, TypeScript, NestJS, and React, with hands-on experience building microservices, payment infrastructure, shipping orchestration, and cloud-native applications on AWS and GCP. Currently backup lead for a 2-4 person engineering team and trusted with independent ownership of new initiatives, with a track record of leading junior developers and applying AI-assisted engineering workflows to deliver reliable, scalable software.

CORE SKILLS

- **Languages:** Node.js, TypeScript, JavaScript, Python
- **Backend:** NestJS, Express, Django, REST APIs, Prisma
- **Frontend:** React.js, Next.js, Angular
- **Databases:** PostgreSQL, MongoDB, Redis, DynamoDB
- **Cloud & DevOps:** AWS, GCP, Docker, CI/CD (GitHub Actions)
- **Architecture:** Microservices, Event-Driven Design, BullMQ, WebSockets, JWT/RBAC
- **AI Tools:** Cursor, Claude, ChatGPT
- **Leadership & Tools:** Team Leading, Code Reviews, Agile/Scrum, Git, Jira, Swagger, Firebase

PROFESSIONAL EXPERIENCE

Senior Software Engineer

Nov 2022 – Present

LMNTO Performance Information Technology (formerly VAI Marketing Management), Dubai

- Owned the end-to-end checkout, payment, and logistics architecture supporting multiple e-commerce platforms and processing hundreds of orders daily.
- Act as backup lead for a team of 2-4 engineers, covering priorities and team direction when the lead is unavailable, and was entrusted with independent ownership of a new initiative outside the lead's involvement.
- Integrated 4 payment gateways, including Stripe, building a modular factory-pattern payment system supporting card and wallet payments.
- Built a shipping orchestration layer that dynamically routes orders to the correct logistics provider based on per-platform configuration and destination-country restrictions, integrating 3 shipping providers with real-time shipment tracking.
- Reduced payment and order-processing failures by building a BullMQ- and Redis-backed retry mechanism with automated notifications triggered when max retries were exceeded.
- Implemented JWT-based authentication with multi-tier role-based access control (RBAC) spanning admin, manager, and staff permission levels, using Prisma as the ORM for the payment and product modules.
- Containerized backend services with Docker and built GitHub Actions CI/CD pipelines that run automated tests and deploy on merge.
- Designed and deployed scalable NestJS microservices powering payments, wallets, and logistics integrations, improving reliability and operational performance.
- Collaborated on React/Next.js integrations across customer and admin platforms.

Software Engineer

May 2021 – Nov 2022

LiveRoom Technologies

- Designed and implemented a real-time auction management admin dashboard (for a live CricFlix auction, UAE) using Firebase Realtime Database, Firebase Auth, and React.
- Led a team of 2-3 junior developers gathering client requirements and delivering enhancements to the Bigpool insurance platform; conducted code reviews for quality and functionality.
- Built AWS Lambda services (Serverless framework) for the Mintzilla NFT marketplace and GRI virtual showroom.

Associate Software Engineer

Jul 2018 – May 2021

LiveRoom Technologies

- Developed REST APIs for the Bigpool insurance platform using Django and PostgreSQL.
- Built a customer web portal and admin dashboard using React; integrated APIs into the React Native mobile app.
- Deployed the Django server on GCP App Engine (GAE) and React apps on GCP Kubernetes Engine (GKE).

EDUCATION

- BSc (Hons) in Computer Science — University College Dublin, Ireland | 2015–2019 (Second-Class Upper Honours, GPA 3.6)

ACHIEVEMENTS

- Star Employee Award — VAI Marketing / LMNTO (2023)

CERTIFICATIONS

- Dubai Centre for Artificial Intelligence & Dubai Future Foundation – One Million Prompts
- DeepLearning.AI – Generative AI with LLMs
- IBM – Fundamentals of Building AI Agents
- Meta – Django Framework
- AWS – Serverless APIs & Applications
- IBM – React Development
- IBM – Agile Development & Scrum
- MongoDB – M150 (Authentication & Authorization), M001 (Basics)